***Optimizing Inventory and Sales Forecasting at Pastry Palace: A Data-Driven Approach to be Efficient and Organised***

**A Final-Term report for the BDM capstone Project**


Submitted by

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## Executive Summary

This project focuses on Pastry Palace, a small-scale bakery that has historically operated on its owner's experience. The business lacked formal systems to track sales and inventory, making it difficult to identify high-performing products, manage perishable goods, and make data-driven decisions to enhance profitability.

To address these challenges, a 50-day dataset (June 1 - July 20, 2025) was manually created through owner interviews and on-site observations. This dataset was analyzed using key methodologies including ABC (Pareto) Analysis, Coefficient of Variation (CV) for sales volatility, Wastage Cost analysis, and a Product Portfolio analysis.

Key findings revealed the 'Fresh' category drives 76% of profit and that the highest financial losses from waste come from high-cost cakes, not just high-volume items. A portfolio analysis categorized products into strategic roles (Stars, Opportunities, etc.), while sales data confirmed predictable weekly patterns. Wastage was traced to specific forecasting errors for high-volume snacks.

Recommendations focus on prioritizing 'Class A' products, implementing a differentiated inventory strategy based on sales volatility, and adopting a portfolio approach to grow high-potential items. The foundational recommendation is the formal adoption of this data tracking system, transitioning the business to a data-driven strategy for sustainable growth.

## 2. Detailed Explanation of Analysis Process/Method

This section outlines the structured approach taken to transform raw, qualitative data into actionable business insights for Pastry Palace. The methodology was designed to be both analytically sound and practically applicable for the business owner, addressing the core problems of inventory control and profitability.

### 2.1 Data Collection and Project Scope

The foundation of this project was the creation of a dataset from scratch due to the absence of a

formal sales or inventory tracking system. The primary goal was to establish a foundational data collection framework and demonstrate the immediate value of data-driven decision-making. This project serves as a powerful **proof of concept**, showing the insights that can be gleaned even from short-term data.

The data collection for the period of **June 1st, 2025, to July 20th, 2025**, was a manual process:

- **Owner Interviews & On-site Observation:** The primary data source was a combination of discussions with the owner and on-site observations to make daily estimations of sales, production quantities, and general profit margins.
- **Manual Stock Count:** For non-perishable items (Packaged, Essentials, and Birthday Items), a manual count of the Initial Stocks was performed daily. This served as a crucial verification point to track inventory movement accurately and identify any discrepancies early on.
- **Data Continuity:** The dataset is continuous across the entire period. Zero sales were recorded for **July 16th**, a day the shop was confirmed to be closed, ensuring a complete and accurate time series for trend analysis.

### 2.2 Data Preprocessing and Structuring

Given the qualitative nature of the data, a critical preprocessing step was undertaken to ensure its integrity and suitability for analysis.

- **Product Categorization:** All items were classified into one of four logical categories to facilitate high-level analysis:
    - **Fresh** (e.g., Pizzas, Cakes, Sandwiches)
    - **Packaged** (e.g., Biscuits, Chocolates, Ice Cream)
    - **Essentials** (e.g., Bread, Milk, Dahi)
    - **Birthday Items** (e.g., Candles, Balloons, Poppers)
- **Item Selection & Exclusion:** To maintain data precision, a decision was made to exclude a few niche items for which verbal estimates were inconsistent. This focused the project on the core products that represent the majority of Pastry Palace's business.

- **Calculation of Key Metrics:** Several key columns were created using formulas to serve as the basis for analysis.
    - *Profit per Unit (₹) = Selling Price (₹) - Cost Price (₹)*
    - *Total Profit (₹) = Profit per Unit (₹) × Quantity Sold*

### 2.3 Analytical Techniques

Each method was chosen specifically to address the project's objectives as outlined in the problem statement.

#### Descriptive Statistical Analysis

- **Method**: A foundational analysis was conducted to summarize the dataset's characteristics using measures like mean, median, and standard deviation for core variables like Quantity Sold and Total Profit.
- **Formula**:
    - Mean:

$$\bar{x} = \frac{1}{n}\sum_{i=1}^{n} x_i$$

    - Median: Middle value of ordered data.
    - Standard Deviation:

$$\sigma = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(x_i - \bar{x})^2}$$

- **Justification**: This directly addresses the core issue of lacking a tracking system by creating the first quantitative baseline of the business's performance.

#### Sales Volatility Analysis (Coefficient of Variation)

- **Method**: The Coefficient of Variation (CV) was calculated for the daily Quantity Sold of each key product.

- **Formula**:

$$\mathrm{CV} = \frac{\sigma}{\mu}$$

- **σ = Standard deviation of Quantity Sold**

  **μ = Mean of Quantity Sold**
- **Justification**: This analysis helps differentiate between products with stable, predictable demand (low CV) and those with erratic, unpredictable demand (high CV). This insight is crucial for creating an effective, differentiated inventory strategy.

**Profit Margin & Wastage Analysis**

- **Method**: This dual analysis involved calculating the specific Profit per Unit (₹) for each item and tracking the Wastage/Remaining column to quantify daily losses.
- **Formula**:
  - Profit per Unit:

    **Profit per Unit=Selling Price−Cost Price**
  - Total Daily Profit:

    **Total Profit=Profit per Unit×Quantity Sold**
  - Wastage Loss:

    **Wastage Loss=Wastage Quantity×Cost Price**
- **Justification**: This directly addresses the need to track item-wise profit and manage perishable stock more effectively.

**ABC Analysis**

- **Method**: Items were classified into A, B, and C categories based on their cumulative contribution to overall profit or revenue, with 'A' items contributing the most.
- **Formula**:
  - Value Contribution per Item:

    **Value=Unit Profit×Quantity Sold**

- Cumulative % Contribution:

  **Calculated as cumulative total of item values as a percentage of the grand total.**

- **Justification**: Helps identify the most valuable items to prioritize for production and stocking.

**Time Series (Trend) Analysis**

- **Method**: By plotting Total Profit (₹) over the 50-day period, recurring trends and weekly cycles were visualized.
- **Formula**:
  - Not a fixed formula, but involves plotting:

    **Profit over Time=f(Date,Total Profit)**
  - Trendline (Linear):

    **A **7-day moving average** was calculated to smooth out daily fluctuations and clarify the underlying trend**
- **Justification**: Identifies consistent demand patterns like weekend surges and midweek drops, helping with proactive stock planning.

**Product Portfolio Analysis**

- **Method:** A scatter plot was used to categorize products into four strategic quadrants (Stars, Workhorses, Opportunities, Dogs) based on their average sales volume versus their profit margin percentage.
- **Justification:** This provides a high-level strategic framework to guide marketing and menu development priorities, helping to increase overall profitability .

**2.4 Tool Justification**

All data management, analysis, and visualization for this project were performed using **Google Sheets**. This tool was chosen for its **accessibility**, **ease of use**, and **sufficient functionality**, ensuring the owner can adopt the system without needing specialized expertise.

## 3. Results and Findings

This section presents the key findings derived from the analysis of the 50-day dataset (June 1 - July 20, 2025). The results are organized to directly address the core business problems of inventory management and profitability. Each finding is presented with a corresponding visualization, followed by a detailed interpretation of its business implications for Pastry Palace.

### 3.1 Finding: High-Value Products Drive Majority of Profit

To pinpoint the specific products responsible for the highest profit contribution, a horizontal bar chart was created to visualize the top 10 performers. The analysis of this chart (Figure 3.1) reveals a significant concentration of profit among a small subset of the shop's freshly prepared items.

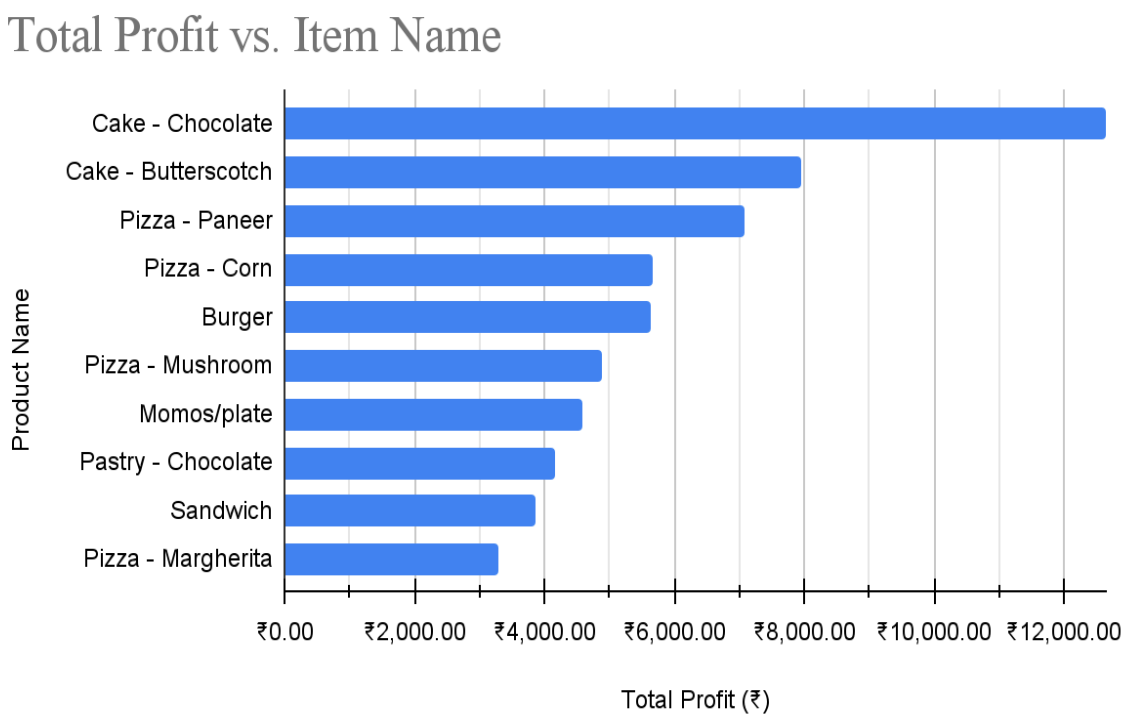


***Figure 3.1: Top 10 Profit-Making Products***

While the chart above identifies the individual top performers, the following summary table provides the broader statistical context of their collective impact on profitability.

| Product group | Number of Items | % of Total Items | Cumulative Profit Contribution |
|---|---|---|---|
| Top 2 Products | 2 | 5.6 | 22.8 |
| Top 10 Products | 10 | 27.8 | 66.0 |
| Class “A” Items | 17 | 47.2 | 81.4 |
| All Products | 36 | 100 | 100 |

A detailed Pareto analysis confirms this trend, revealing that the top 17 products, representing just under half (47%) of the product variety, generate over **81% of the total profit**. This highlights a group of 'vital few' products that are critical to the business's financial health.

The most profitable items are overwhelmingly from the **'Fresh'** category. The two top-performing products alone, **Cake - Chocolate** and **Cake - Butterscotch**, contribute a combined **22.8% of the total profit from all items**. As shown in the chart, these are followed by high-margin fresh items like Pizza - Paneer, Pizza - Corn, and Burger. This strong performance from the 'Fresh' category underscores its importance as the primary engine of profitability for Pastry Palace.

**Business Implication:** This analysis provides a clear, data-driven directive for inventory management and sales strategy. The identified "Class A" products are mission-critical. The owner's top priority should be to ensure their consistent availability and quality. These high-value items are also the best candidates for promotional activities to further maximize revenue.

**Finding 3.2: 'Fresh' Category is the Dominant Profit Center**

To understand the business's performance at a broader, strategic level, a pie chart was generated to illustrate the profit contribution of each product category. This high-level view (Figure 3.2)

clearly shows that the business's financial success is overwhelmingly driven by one dominant category.

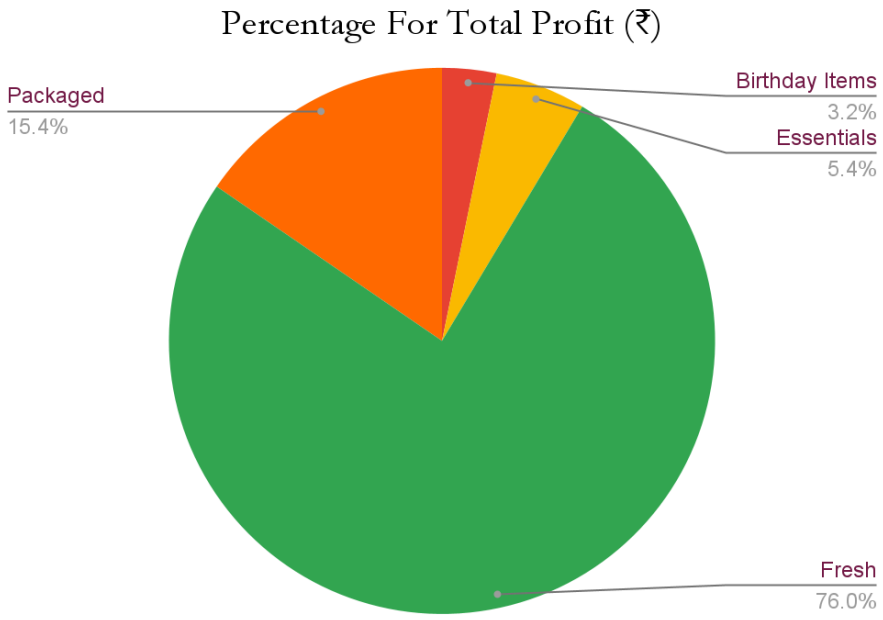


***Figure 3.2: Profit Contribution by Category***

The analysis reveals a clear hierarchy of profitability among the categories:

- The **'Fresh'** category alone accounts for **76%** of the total profit.
- The **'Packaged'** goods category is the second-largest contributor, bringing in **15.5%** of the profit.
- **'Essentials'** (5.4%) and **'Birthday Items'** (3.2%) make up the remainder, playing a smaller, supplementary role in the business's revenue stream.

This category-level insight provides crucial context for our previous product-level analysis. The dominance of the 'Fresh' category is a direct result of the high performance of the 'Class A' items like cakes, pizzas, and burgers, which all fall under this classification.

**Business Implication:** This finding strongly suggests that the core of the business's value lies in its freshly prepared **food**. Strategic decisions related to growth, menu development, and marketing should be centered on the 'Fresh' category. While other categories are important for

providing a complete customer experience, the primary path to increased profitability is through the optimization and potential expansion of **these high-margin** fresh food offerings.

**Finding 3.3: Sales Volatility Analysis Identifies Predictable and Erratic Products**

To enhance the inventory management strategy, an analysis was conducted to understand the predictability of sales for different products. For this, the Coefficient of Variation (CV) was calculated for the daily **Quantity Sold** of each key item. The CV measures how volatile sales are relative to their average; a lower CV indicates stable and predictable demand, while a high CV indicates erratic and unpredictable demand.

| ***Item Name*** | ***Average Daily Sales*** | ***Coefficient of Variation (CV)*** |
|---|---|---|
| ***Most Volatile Products (High CV)*** | | |
| Birthday - Poppers | 1.02 | 65.69% |
| Birthday - Caps | 2.44 | 63.93% |
| Dry Fruit - Kaju | 1.36 | 59.33% |
| Birthday - Candles | 1.86 | 53.62% |
| Cake - Butterscotch | 1.54 | 50.57% |
| ***Most Stable Products (Low CV)*** | | |
| Chocolate - ₹40 Bar | 5.14 | 30.00% |
| Ice Cream - Amul | 8.38 | 25.49% |
| Pav Bhaji | 5.91 | 21.97% |
| Sandwich | 7.44 | 19.76% |
| Momos/plate | 10.60 | 19.62% |

The analysis reveals two distinct groups of products with very different sales patterns:

- **High Volatility (Unpredictable) Products:** The items with the highest CV are primarily from the 'Birthday Items' category, such as Birthday - Poppers (CV 65.7%) and Birthday - Caps (CV 63.9%). Their demand is erratic and event-driven, characterized by many days of zero sales followed by small, unpredictable purchases. This makes their daily demand very difficult to forecast accurately.
- **Low Volatility (Predictable) Products:** In stark contrast, core 'Fresh' snack items have the most stable and predictable sales. The analysis reveals that **Momos/plate** is the most consistent seller with the lowest CV of just **19.6%**, closely followed by **Sandwich** at **19.8%**. The newly consolidated Pav Bhaji item also ranks among the most stable products with a low CV of 22.0%.

  Notably, **Momos/plate** is now confirmed to be the ideal product in the portfolio, combining the **highest average daily sales (10.6 units)** with the **lowest sales volatility**, making it a reliable and high-volume profit driver.

**Business Implication:**

This volatility analysis provides a crucial layer of insight for developing a more sophisticated inventory strategy, directly addressing the goal of managing stock and avoiding stockouts.

- **For high CV items:** A standard daily stock level is inefficient. A "safety stock" approach is more appropriate to cater to unexpected demand without tying up capital.
- **For low CV items:** The owner can have high confidence in sales forecasts. This allows for lean inventory management, minimizing both stockouts and financial losses from over-preparation.

**Finding 3.4: Time Series Analysis Reveals Weekly Patterns and Event-Driven Spikes**

To understand the business's performance over time and identify temporal patterns, the daily profit was plotted for the entire 50-day period The analysis of the daily profit trend reveals several key insights into the operational rhythm of the business:

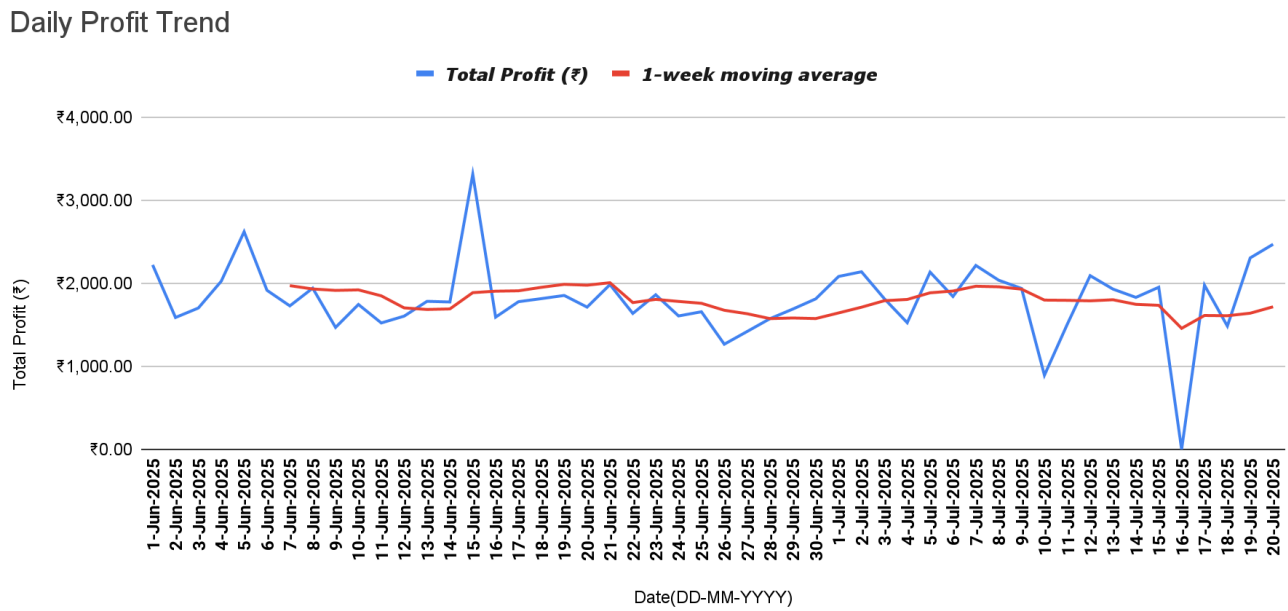


***Figure 3.3: Daily Profit Trend (June 1 - July 20, 2025)***

- **High Sensitivity to Events:** The most significant feature is the business's response to specific events. A major spike in profit to **₹3,317.00** occurred on **June 15th**, coinciding with Father's Day, which drove high sales of celebratory items like cakes. A smaller but distinct peak is also visible around **June 21st** (**₹1,987.25**), which can be attributed to a local office event.
- **Frequent Friday Dips:** The data shows a frequent, though not universal, trend of profit dips on Fridays. This pattern supports the owner's observation that the nearby local market can draw foot traffic away on that day.
- **Operational Realities:** The data captures operational anomalies, such as the sharp drop to zero profit on **July 16th**, the day the shop was closed.
- **Stable Overall Trend:** Despite the significant daily volatility, the business's overall profitability remained relatively stable across this 50-day period, with no clear long-term upward or downward trend.

**Business Implication:**

This analysis is crucial for improving **staff scheduling, inventory planning, and promotion timing**. The clear impact of holidays demonstrates a strong opportunity to plan ahead by increasing stock of high-margin items before known events. The tendency for a "Friday dip" allows the owner to be more cautious with the preparation of fresh items on that day, helping to reduce potential wastage.

**Finding 3.5: Wastage Analysis Pinpoints Specific Products for Cost Reduction**

To directly address the problem of financial losses from unsold items, an analysis was conducted to identify which specific 'Fresh' products contributed the most to wastage.

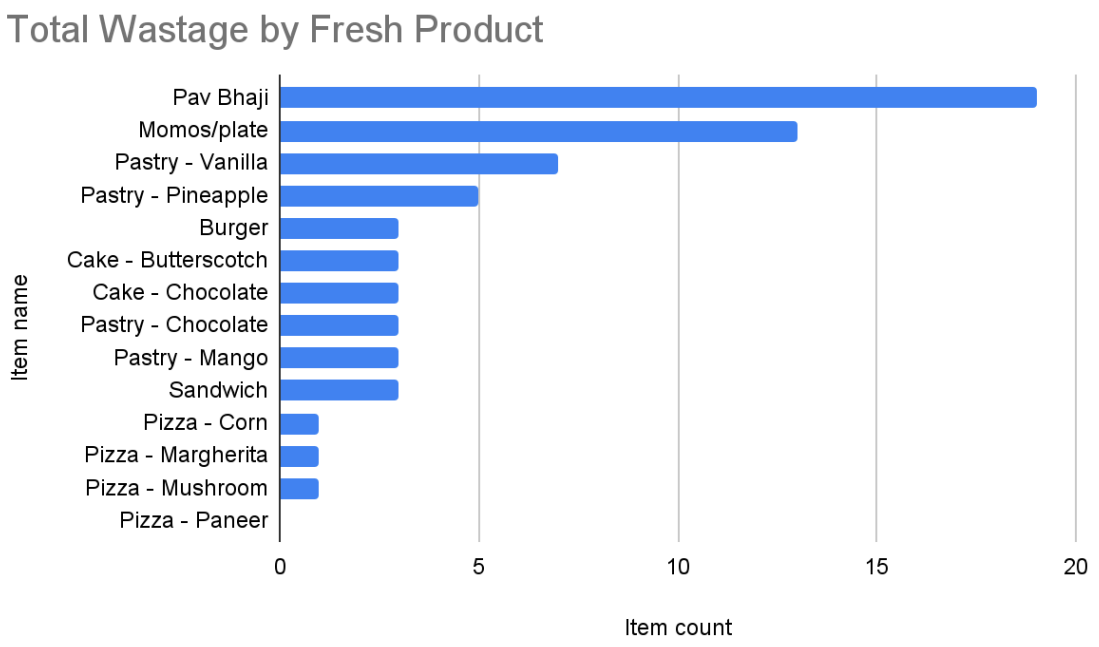


***Figure 3.4: Total Wastage by Fresh Product (June 1 - July 20, 2025)***

- The analysis of the wastage data reveals that the majority of losses are concentrated in a few specific snack items and are primarily caused by **forecasting errors rather than operational issues**. After consolidating inconsistent naming in the raw data, **Pav Bhaji** emerges as the single largest source of waste with a total of **19 units** discarded. The second-highest is **Momos/plate**, with **13 wasted units**.

- In stark contrast, the pizza line was managed with exceptional efficiency. Four different pizza varieties had only one unit of waste or less over the entire 50-day period, with the high-value Pizza - Paneer recording zero waste.
- **Business Implication:**
- This finding provides a highly targeted path to cost reduction. The primary focus for waste reduction should be on **Pav Bhaji** and **Momos/plate**, which together accounted for **32 units of waste** in the analysis period. Refining the daily preparation quantity for just these two dishes will have the most significant impact on the bottom line. The successful, low-waste management of the pizza line can serve as a best-practice model for other fresh items.

**Finding 3.6: Financial Analysis of Wastage Reveals High-Cost Items as Top Source of Loss**

While the previous wastage analysis identified which products were wasted most frequently, this analysis was conducted to determine which products were causing the largest **financial drain** on the business. To do this, the Wastage Cost was calculated for each fresh item (Wastage Quantity × Cost Price).

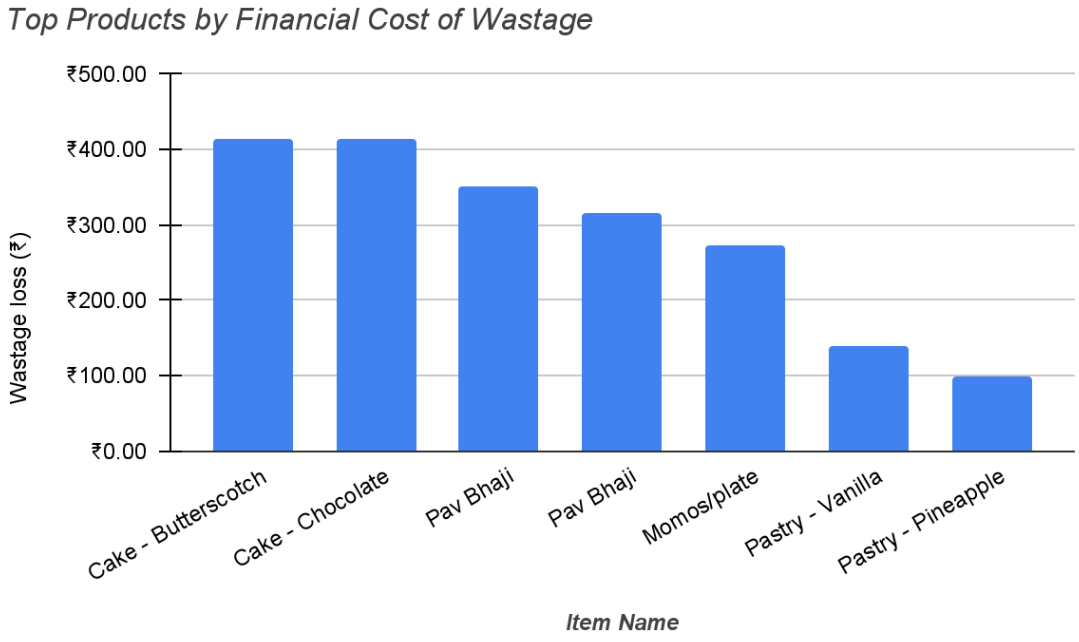


***Figure 3.5:Top 7 Products by Financial Cost of Wastage***

The results of this financial analysis are striking and reveal a new perspective on waste management. The two products responsible for the highest financial losses are **Cake - Butterscotch (₹414.00)** and **Cake - Chocolate (₹414.00)**.

This is a critical insight because these items did not appear as the top sources of waste when measured by unit count alone. A single unsold cake, with its high cost price of ₹138, represents a financial loss equivalent to roughly seven unsold plates of Momos. Following the cakes, the high-volume snack items **Pav Bhaji** (combined total of ₹665) and **Momos/plate** (₹273) are also significant sources of financial loss.

**Business Implication:**

While reducing waste on frequently unsold items like Momos is important, preventing the loss of even a single high-cost cake has a much larger positive impact on the daily profit margin. This reinforces the recommendation to consider a **pre-order model for specialty cakes**, as this would almost entirely eliminate the single largest source of financial loss from waste.

**Finding 3.7: Product Portfolio Analysis Reveals Strategic Categories**

To develop a higher-level business strategy beyond individual metrics, a portfolio analysis was conducted. This analysis categorizes products based on a combination of their average sales volume and their profit margin percentage, providing a roadmap for marketing and inventory focus.

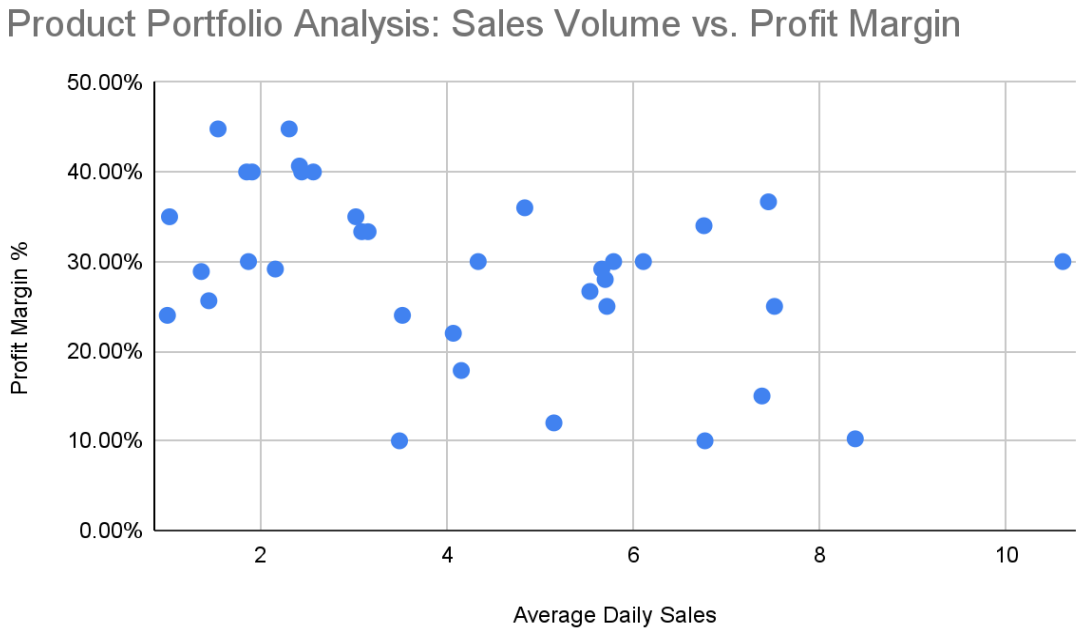

***Figure 3.6: Sales Volume vs. Profit Margin Analysis***

The analysis positions each product into one of four strategic quadrants, providing a clear view of its role in the business:

- **Stars (High Volume, High Margin):** These are the ideal products that are both popular and highly profitable per unit. The key items in this category are core 'Fresh' snacks like **Momos/plate** (10.6 avg. daily sales, 30.0% margin) and **Sandwich** (7.4 avg. daily sales, 36.7% margin).
- **Workhorses (High Volume, Low Margin):** This group includes products that sell frequently but do not offer high margins. They are critical for generating consistent daily revenue. Key examples are **Ice Cream - Amul** (8.4 avg. daily sales, 10.2% margin) and low-price chocolates like the **Chocolate - ₹10 Bar** (6.8 avg. daily sales, 10.0% margin).
- **Opportunities (Low Volume, High Margin):** This crucial quadrant contains the shop's most profitable items on a per-unit basis, but they do not yet sell in high volumes. The most prominent examples are **Cakes** (44.8% margin) and **Pizzas** (40% margin), which have average daily sales between 1.5 and 2.6 units. These items represent a significant opportunity for growth.
- **Dogs (Low Volume, Low Margin):** This quadrant contains items that contribute minimally to both sales volume and profit, such as some of the less popular packaged goods.

**Business Implication:**

This portfolio analysis provides a powerful strategic framework. The business should:

1. **Protect and Promote its "Stars"** to maintain their strong performance.
2. **Efficiently Manage its "Workhorses"** to ensure they are always in stock to drive revenue.
3. Focus marketing efforts on **"Opportunities,"** aiming to increase the sales volume of these high-margin products and turn them into future Stars.
4. **Evaluate its "Dogs"** to determine if they should be replaced with new, potentially higher-performing products.

## 4. Interpretation of Results and Recommendations

### 4.1 Overall Interpretation

The analysis of the 50-day dataset provided a clear and data-driven picture of Pastry Palace's operations. The findings confirm that the business's profitability is heavily concentrated in a small number of 'Fresh' products. Sales are influenced by predictable weekly patterns and specific events, and there are clear opportunities to reduce costs by addressing wastage in a few key items. The following recommendations provide a roadmap for the owner to leverage these insights to improve inventory control and increase profitability.

### 4.2 Actionable Recommendations

#### 4.2.1 Recommendation: Focus on Core Profit Drivers

- **Short-Term (Next 4 Weeks):**
  - **"Must-Stock" Policy:** It is recommended that the owner implement a priority policy for the **Top 10 'Class A' products**, with the goal of achieving a near-0% stockout rate for these items.
  - **In-Store Promotion:** Consider creating a simple "Our Bestsellers" sign featuring the top 3-5 products to guide new customers to high-margin items.
- **Long-Term (Next 3-6 Months):**
  - **Menu Optimization:** It is advised to review the performance of the bottom 5-10 **'Class C' products**. If an item has low sales and low profitability, consider removing it from the menu to simplify inventory and operations.
  - **Supplier Negotiation:** Use the sales data to show suppliers the high volume of key ingredients purchased for top 'Fresh' items and attempt to negotiate better pricing.
- **Potential Impact:** Focusing resources on high-value products will ensure the business consistently satisfies demand for its most profitable items, leading to increased customer loyalty and higher overall profit margins.

#### 4.2.2 Recommendation: Implement a Differentiated Inventory Strategy

- **For High-Volatility (High CV) Products (e.g., Birthday Poppers, Cake - Butterscotch):**
    - **Short-Term (Next 4 Weeks):** Establish a **minimum stock policy**. Maintain a small, fixed quantity on the shelf (e.g., 5-10 units for birthday items) and reorder only when stock falls below a set point (e.g., 3 units).
    - **Long-Term (Next 3-6 Months):** For high-cost, high-CV items like Cake - Butterscotch, introducing a **pre-order model** can reduce waste to zero and guarantee sales.
- **For Low-Volatility (Low CV) Products (e.g., Momos/plate, Sandwich):**
    - **Short-Term (Immediately):** Use the average daily sales data from this report to set reliable daily production targets. For Momos/plate (with average sales of 10.6 units), the owner can confidently prepare a consistent number each day (e.g., 11-12 units).
- **Potential Impact:** This differentiated strategy will make inventory management far more efficient, reducing capital tied up in slow-moving items and cutting down on food waste from over-preparing daily staples.

#### 4.2.3 Recommendation: Align Operations with Predictable Sales Trends

- **Short-Term (Immediately):**
    - **Friday Production Adjustment:** On Fridays, proactively reduce the preparation quantity of high-wastage 'Fresh' items (like **Pav Bhaji** and **Momos**) by 15-20% compared to the weekly average.
    - **Create an Event Calendar:** Keep a simple calendar of local holidays and events to anticipate high-demand days at least a week in advance.
- **Long-Term (Next 3 Months):**
    - **Event-Specific Promotions:** For major events, consider creating special combo offers (e.g., 'Cake + Birthday Banner') and track their success to refine future promotions.
- **Potential Impact:** Aligning production with known sales patterns will directly reduce wastage costs, while proactive planning for events will maximize revenue opportunities.

**4.2.4 Recommendation: Implement Targeted Measures to Reduce Wastage** The analysis pinpointed forecasting errors for high-volume snacks and the high financial impact of unsold cakes as the primary drivers of loss.

- **To Address Forecasting Waste (for Pav Bhaji and Momos):**
    - **Immediate Action:** It is recommended to implement a daily **"Wastage Log."** At the end of each day, record the number of unsold Pav Bhaji and Momos units. After one week, this log should be compared to the sales data to adjust the following week's preparation quantity down.
- **To Address High-Cost Waste (for Cakes):**
    - **Short-Term:** Consider a "High-Value Item" End-of-Day Policy. If a whole cake is unsold near closing time, offering it at a significant discount (e.g., 50% off) can help recover the high ingredient cost.
- **Potential Impact:** These targeted recommendations will have a direct and immediate impact on the business's bottom line by reducing the cost of goods sold.

**4.2.5 Recommendation: Adopt a Portfolio Strategy for Growth** The portfolio analysis provides a strategic guide for growth.

- **For "Stars" (e.g., Momos, Sandwich):**
    - **Action: Protect and Maintain.** Ensure quality is high and they are always in stock.
- **For "Workhorses" (e.g., Ice Cream - Amul):**
    - **Action: Manage for Efficiency.** Focus on efficient ordering to prevent stockouts.
- **For "Opportunities" (e.g., Cakes, Pizzas):**
    - **Action: Develop and Grow.** Actively try to increase the sales volume of these high-margin items through promotions and in-store suggestions.
- **For "Dogs" (Low-profit, low-sales items):**
    - **Action: Evaluate and Simplify.** Review these products quarterly and consider removing them to simplify operations.

- **Potential Impact:** Adopting this portfolio strategy will help the business to be more strategic, focusing its limited resources on the products with the highest potential to grow profits.

**4.2.6 Foundational Recommendation: Formalize and Continue Data Collection** The most critical recommendation is to build upon the foundation created by this project.

- **Immediate Action:** Formalize the spreadsheet-based tracking system as a **permanent daily business process**.
- **Long-Term Goal (Next 12 Months):** Continue this data collection for a full year to reveal **seasonal trends**. This long-term data will be invaluable for more accurate forecasting and strategic menu planning.
- **Potential Impact:** A continuous data stream will transform decision-making from reactive intuition to proactive, data-driven strategy, enabling sustained growth and improved efficiency.

This project successfully transitioned Pastry Palace from an intuition-based operational model to a data-driven one. By creating the business's first-ever structured dataset and applying targeted analytical techniques, this study provided a clear, quantitative picture of the shop's performance. The analysis identified the mission-critical 'Fresh' products that form the financial backbone of the business, quantified sales volatility to enable a more sophisticated inventory strategy, and pinpointed specific causes of wastage that can be immediately addressed to reduce costs. The findings and subsequent recommendations provide a clear, actionable roadmap for the owner to enhance profitability, improve efficiency, and reduce waste. Ultimately, this project serves as a powerful proof of concept that even simple, consistent data tracking can unlock significant value and create a foundation for sustainable growth for a small business.

**5. Limitations of the Project**

- **Limited Data Duration:** The analysis is based on a 50-day period. While sufficient for this pilot study, it is too short to identify and validate long-term seasonal trends.

**6. Links to Project Resources - 📂 final term**